Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

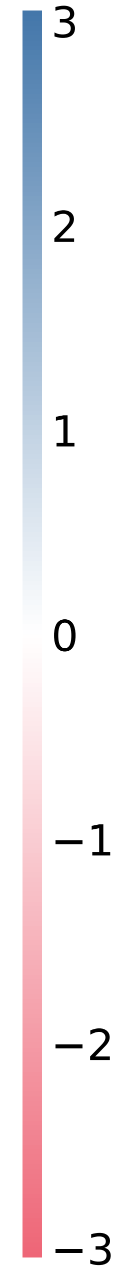
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think> To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV, we need to use the de Broglie wavelength equation Î» = h / âĪļ(2mK), where h is the Planck's constant, m is the mass of the electron, and K is the kinetic energy. First, convert the kinetic energy from eV to J: 1 eV = 1.6 ÃĹ 10^-19 J. Then, use the Planck's constant (h = 6.626 ÃĹ 10^-34 Js) and the mass of the electron (m = 9.11 ÃĹ 10^-31 kg) to calculate the de Broglie wavelength.</think><answer>8.26 ÃĹ 10^-9 m / 100</answer>